Michael J. Ledford

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Education

University of Maryland, College Park — *College Park, MD*

Doctor of Philosophy: Computer Science — *August 2024 – Expected August 2027*

- Advised by Dr. William (Bill) Regli
- Focus: Artificial Intelligence/Machine Learning, Human-Computer Interaction

Air Force Institute of Technology — *Dayton, OH*

Master of Science: Electrical Engineering — *August 2015 – March 2017*

- Specialization: Digital Engineering
- Advisor: Richard Martin
- Thesis: Development of a Software-Defined GNSS Simulation Prototype For Advanced Signals Research
- GPA: 3.49

North Carolina State University — *Raleigh, NC*

Bachelor of Science: Computer Engineering, Electrical Engineering — *August 2010 – May 2015*

- Minor in *Aerospace Studies*
- GPA: 3.78

Academic Experience

Air Force Institute of Technology Autonomy & Navigation Technology (Center) — *Dayton, OH*

Graduate Researcher — *November 2015 – March 2017*

North Carolina State University – Innovation & Entrepreneurship — *Raleigh, NC*

Social Entrepreneur — *August 2014 – May 2015*

Publications

Ledford, Michael J., "*Development of a Software-Defined GNSS Simulation Prototype for Advanced Signals Research.*"
Faculty Advisor: Dr. Richard K. Martin.
Sponsor: Air Force Research Lab: Space Vehicles Directorate (AFRL/RV). [ANT]
URL: https://apps.dtic.mil/sti/tr/pdf/AD1054687.pdf

P. Patel, **M. Ledford**, S. Gunawardena, "*Development of a GNSS Waveform Prototyping Platform for Advanced Signals Research,*" Proceedings of the 2017 Joint Navigation Conference of the Institute of Navigation, Dayton, Ohio, June 2017.

Honors & Awards

2023 Academic Year 2024 Dr. Heather Wilson STEM PhD Fellowship
2022 Wing Company Grade Officer (CGO) of the Year
2022 Wing Staff Company Grade Officer (CGO) of the Year
2021 Wing Operations Flight Commander of the Year
2021 Joint Service Commendation Medal
2021 Operation Inherent Resolve Campaign Medal
2019 Squadron Company Grade Officer (CGO) of the Year

2018 Air Force Achievement Medal with Oak Leaf Cluster
2017 Chief of Staff of the Air Force (CSAF) Star Performer
2015 Summa Cum Laude, North Carolina State University
2015 Distinguished Graduate, Air Force Reserve Officer Training Corps
2007 Eagle Scout (Boy Scouts of America)

Scholarships

2023 Academic Year 2024 Dr. Heather Wilson STEM PhD Program Scholarship

Talks & Presentations

May 2016. *GPS Waveform Prototyping Platform (GWPP) Demystified.* Presented to Director, Sensors Directorate Air Force Research Lab

Professional Memberships

Association for the Advancement of Artificial Intelligence (AAAI)
Association of Old Crows (AOC)
Eta Kappa Nu (HKN)
Institute of Electrical and Electronics Engineer (IEEE)

Military Experience

Officer, Major, United States Air Force (USAF) — *August 2015 - Current*

Developmental Engineer / Acquisition Officer

350th Spectrum Warfare Wing (350 SWW) — *Eglin AFB, FL*

Deputy Chief, Commander's Action Group — *May 2022 – August 2024*

- Provided executive support, strategic advice, and critical project support to the Wing Commander
- Assisted the Wing Commander in formulating strategic messaging and executive engagement plan

36th Electronic Warfare Squadron (36 EWS) — *Eglin AFB, FL*

Flight Commander, Electronic Warfare — *May 2020 – May 2022*

- Oversaw the development, testing, and fielding of Electronic Warfare combat mission software
- Led 34-member team to execute $2.8M Electronic Warfare Integrated Reprogramming portfolio

Air Force Research Laboratory (AFRL) — *Wright-Patterson AFB, OH*

Mobile Application Developer — *August 2017 – March 2018*

- Developed first Android-based mobile application to request and track equipment resupply
- Integrated application into Android Tactical Assault Kit (ATAK) geospatial infrastructure and fielded to special operations forces operators worldwide

National Air and Space Intelligence Center (NASIC) — *Wright-Patterson AFB, OH*

Officer in Charge, Enterprise Web Development / Fiber Optics Analyst — *April 2018 – May 2020*

- Defined, designed, and developed NASIC's web enterprise in support of global stakeholders across intelligence, acquisition, and warfighter communities
- Researched and identified critical adversarial fiber optics networks essential to command and control operations

Industry Experience

Qualcomm — *Raleigh, NC*

CPU Design Verification Intern & Contractor — *August 2014 – August 2015*

- Created Perl script to enhance validation and documentation for System Verilog source code
- Conducted Simulation performance analysis
- Performed coherent bus protocol regression testing of partial good multiprocessor model configuration

Intel Corporation — *Hillsboro, OR*

Full Chip Integration and Validation Intern — *May 2014– August 2014*

- Created Perl and Python scripts to enhance post-silicon integration validation of a security engine IP into a next-generation System on a Chip (SoC)
- Developed low-level firmware to initialize memory to support the development of a new post-silicon validation tool
- Assisted in creating a Verification Test Plan for the integration of an updated security engine IP

Cisco Systems — *Research Triangle Park, NC*

Hardware Developmental Engineer Intern — *May 2013 – August 2013*

- Performed Design Verification Testing for the ASR1000-ESP200 prototype
- Created automation testing scripts using Perl to test hardware components on motherboards and daughtercards on next-gen routers
- Upgraded engineering chassis prototypes for emerging product lines

Ideal Fastener Corporation — *Oxford, NC*

Electrical Engineer Intern — *May 2012 – August 2012*

- Used Draftsight software to draw electrical circuits for zipper machines and panel layouts on the production floor
- Designed and constructed electrical circuits for emerging zipper machines

Skills & Related Course

Skills

Languages	Python, Java, Dart, Go, MATLAB, Dart
Framework	PyTorch, TensorFlow, TensorFlow-Lite, Android, Flutter, Docker, Kubernetes, Django

Related Courses

AI/ML	Udemy: AI A-Z; Machine Learning A-Z; Deep Learning A-Z; TensorFlow Developer Coursera: Supervised Machine Learning: Regression and Classification Natural Language Understanding and Common-Sense Reasoning
Theory	Design and Analysis of Algorithms; Design Principles of Computer Architecture
Math	Multivariable Calculus; Practical Statistics for Data Scientists; Linear Algebra; Differential Equations
Security	Human Factors in Security and Privacy

Extracurricular Activities

Flutter Mobile Application Developer

January 2020 – Current